



Preliminary Data from Demographic and Health Surveys on Infant Feeding in 20 Developing Countries^{1,2}

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Abstract

This study describes infant feeding practices in developing countries, specifically complementary liquids and foods in the first year of life. Data were compiled from Demographic and Health Surveys conducted from 1999 to 2003. We analyzed data from those countries with available data, including results for child-level 24-h and 7-d food and fluid intakes. We used datasets from 20 countries with information on >35,000 infants categorized by age: 0–6 and 6–12 mo. For analysis, we grouped data for fluids other than breast milk as water, other milk (e.g., tinned, powdered, animal), infant formula, and other liquids (e.g., fruit juice, herbal tea, sugar water). All specific solid foods were grouped as any solid foods. We present data on breast-feeding and maternal-reported fluid and solid intake by infants in a 24-h period, for individual countries, and in a pooled analysis. Pooled data show that 96.6% of 0- to 6- and 87.9% of 6- to 12-mo-old infants were currently breast-fed. Reported feeding of other fluids was lower among 0- to 6-mo-olds than 6- to 12-mo-olds: water (45.9 vs. 87.4%), other milk products (11.9 vs. 29.6%), infant formula (9.0 vs. 15.1%), and other liquids (15.1 vs. 41.0%). Pooled analysis showed that 21.9% of mothers reported feeding 0- to 6-mo-old infants some type of solid food, and 80.1% of mothers reported feeding solids to 6- to 12-mo-olds. These survey data show that other milks, other liquids, and solid foods are each much more commonly fed throughout infancy than commercial infant formulas in the countries studied. *J. Nutr.* 137: 518S–523S, 2007.

Introduction

Infant research in developing countries has focused on understanding the relations among the complex factors affecting infant health to help guide policy within individual countries and globally (1–5). Among such factors, breast-feeding is an important focus for research and policy related to infant morbidity

and mortality; recent papers of Bahl et al. (6) and Bhandari et al. (7) emphasize these issues. To improve infant health, we need a better understanding of the timing, use, and role of formulas, nonformula liquids, and complementary foods during the first year of life and of the familial decisions that determine these infant-feeding patterns.

In this article, we describe the intake of infants in 20 developing countries using secondary analysis of data from the Demographic and Health Surveys (DHS). The purpose of this article is to describe the study population and initial results of reported feeding practices of infants. The foci of our analysis were the responses of mothers when asked whether they were currently breast-feeding and whether they had fed other liquids and solid foods to their infants in the previous 24 h.

Methods

Dataset. DHS have been conducted since 1986, originally focusing on contraception and family planning. The purpose of DHS was expanded to provide a standardized survey instrument that can be used by developing countries to collect data on maternal/infant health, intake, and household variables and to build national health statistics (8). DHS currently include nationally representative samples of women of childbearing age. The limitations of DHS include reporting and recall bias, a problem common to other questionnaires, particularly for retrospective data relying on memory of past events (9).

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Under contract with the U.S. Agency for International Development since 1989, Macro International, Opinion Research Corporation (ORC Macro) updates the DHS basic model instrument approximately every 5 y and provides assistance to in-country agencies and institutions. The DHS core questionnaire is designed to collect information on anthropometry, child feeding, and child morbidity. Specific questions and answers are modified to be country or culturally specific. Countries may select to run surveys multiple times or once. Since the inception of the DHS in 1984, 76 countries globally have participated in surveys.

ORC Macro has implemented the DHS to provide standardization from country to country in terms of quality of data collection and data analysis. Because of the use of uniform DHS instruments, cross-country comparisons and data pooling are possible to study child feeding and other factors that have a bearing on health outcomes in the developing countries that participate in the DHS (9–11).

Samples. For the purpose of this study, we accessed MEASURE DHS+ data collected from developing countries between 1999 and 2003, using the Model B survey instrument. Only 24-h and 7-d intake questions were included in the DHS survey instruments between 1999 and 2003. During this period, 38 surveys were conducted in 5 global regions. The number of countries with surveys during the time period examined are listed with the total number of countries where surveys have ever been conducted in these global regions as follows (number during time period/total number): Sub-Saharan Africa (20/37), North Africa/West Asia/Europe (4/9), South Asia/Southeast Asia (7/11), Central Asia (2/4), and Latin America/Caribbean (5/15). We selected for the study only those countries that provided child-level focused datasets, had included 24-h and 7-d intake questions, and had data that had been made available by the country, including an English-language survey instrument. For our preliminary analysis we excluded the Latin America/Caribbean region. Our data sources therefore included (number of countries of number during time period): 9 of 20 in Sub-Saharan Africa [Ethiopia (2000), Ghana (2003), Kenya (2003), Malawi (2000), Namibia (2000), Nigeria (2003), Uganda (2001), Zambia (2001), Zimbabwe (1999)]; 3 of 4 in North Africa/West Asia/Europe [Armenia (2000), Egypt (2000), Jordan (2002)]; 7 of 7 in South Asia/Southeast Asia [Bangladesh (1999), Cambodia (2000), India (1999), Indonesia (2002), Nepal (2001), Philippines (2003), Vietnam (2002)]; 1 of 2 in Central Asia [Kazakhstan (1999)]; and 0 of 5 in Latin America/Caribbean. The Eritrea (2002) and Turkmenistan (2000) data that met our inclusion criteria had not been released by the countries when we began our study.

We verified that the data we downloaded were the correct dataset for our analyses. We also calculated the country-specific sample numbers and compared these numbers to reports on the ORC Macro website to verify that full datasets were included for each country. For our study, we selected the youngest child less than 1 y of age in each of the families, if the youngest child was living. We excluded deceased children and all children for whom age information was missing. Multiple children from 1 mother were included in the analysis only if the children were born at the same time (i.e., twins or triplets). Analyses were conducted separately for infants <6 mo old (0–6 mo) and infants 6 mo or older but <12 mo old (6–12 mo). For the 20 countries, we used data from 35,648 infants (18,174 aged 0–6 mo and 17,474 aged 6–12 mo). The number of infants sampled in each country ranged from as few as 99 (0–6 mo) and 115 (6–12 mo) in Kazakhstan to 5533 (0–6 mo) and 5083 (6–12 mo) in India. For the 2 age groupings, the mean age (SD) was 2.7 (0.1) and 8.5 (0.1) months for 0- to 6-mo-old and 6- to 12-mo-old infants, respectively.

Variables. From the Model B Women's Questionnaire for 1999–2003, we used the following variables for this descriptive analysis: maternal last birth, next-to-last birth, child living or deceased, child twin, child age, breast-feeding still, times breast-fed in last 24 h, other liquids in last 24 h, solid foods in last 24 h, liquids in the last 7 d, and solids in last 7 d. For this cross-country comparison, we included those countries where an English language questionnaire was provided so we could carefully review each question set for skip patterns and country-specific variations in question wording or reply options. For each country, we reviewed coding of variables to confirm comparability among the 20 countries. Several countries introduced small wording changes into the survey

questions; more frequently, cross-country variability was introduced by the addition or substitution of response options. Additional cross-country variation occurred because some countries chose to omit questions in the model survey. Thus, the numbers of countries providing responses are fewer for some variables than other variables. Among the variables presented in this article, Namibia did not inquire whether infants were fed tinned, powdered, or other milks; Ethiopia, Namibia, India, and Nepal did not inquire if infants were fed commercial formula.

One inclusion criterion for this analysis was that the model questionnaire used by the country included both 24-h and 7-d intake questions for liquid and food. Some of the countries that used these 2 questions, however, did not include both questions for all liquid or food types. In 4 countries (Jordan, Indonesia, Nepal, and the Philippines) a negative response to a food or liquid type on the 7-d intake question resulted in a missing data code in the parallel 24-h question. The other 16 countries instead recorded a zero value. We recoded all 24-h liquid and food intake variables so that if the 7-d variables for an item had a value of zero, then the 24-h variable also had a zero value.

For this analysis, food and liquid variables other than water and infant formulas were collapsed into broader food and liquid categories to permit cross-country comparisons. Specifically, we collapsed tinned, powdered, and any other specified animal milks; all other liquids such as fruit juices, herbals teas, and sugar water (excluding unique country-specific liquids) into other liquids, and the 10 types of solid food groups into an any-solid-foods category.

Analytical methods. In the model questionnaire, mothers were asked if they were still breast-feeding and how many times their infant was fed specific liquids or solids in the last 24 h. To compute the percentage of infants who consumed items, we recoded the variables so that any response indicating consumption was set to 1 (consumed item), and any response indicating no consumption or didn't know whether there was consumption was set to 0 to indicate no consumption. Missing values were treated as missing. Mean values are calculated across the 20 countries.

We used sample weights for all analyses. DHS+ data are weighted by country to account for variability of the country-based sampling models overall. In countries where only married women were sampled, we adjusted the rates to reflect all women in the country, and we standardized the weights for individual countries to represent the number of respondents on a variable basis.

We also pooled data from the 20 countries to provide a large sample size, increase statistical power, and present a more global perspective. According to Hatt and Waters (10), pooling can reveal broader results that are "often obscured by the noise of individual data sets." To calculate the pooled values an additional adjustment is needed to account for the variability in the number of individuals sampled in each country. This is accomplished using the weighting factor $1/(A \cdot n_c/n_t)$, where A is the number of countries asked a particular question, n_c is the number of respondents for the country c , and n_t is the total number of respondents over all countries asked the question (11).

Results

Figures 1 and 2 provide the percentage of infants whose mothers reported that they were still breast-feeding and the percentage of infants who were fed 4 categories of liquids and any solid food in the last 24 h for 0- to 6- and 6- to 12-mo-olds, respectively. Table 1 presents the pooled summary of these data for the period 1999–2003.

Current breast-feeding averages for these 20 countries ranged from 80.5% (Philippines) to 100% (Malawi, Zimbabwe, Nepal) of women who were asked this question for infants aged 0–6 mo. Percentages of current breast-feeding were lower for infants aged 6–12 mo, ranging from 53.4% in Armenia to 99.4% and 99.5% in Malawi and Zambia, respectively. Reported current breast-feeding was consistent within countries across the 2 age categories in that countries whose mothers reported the highest

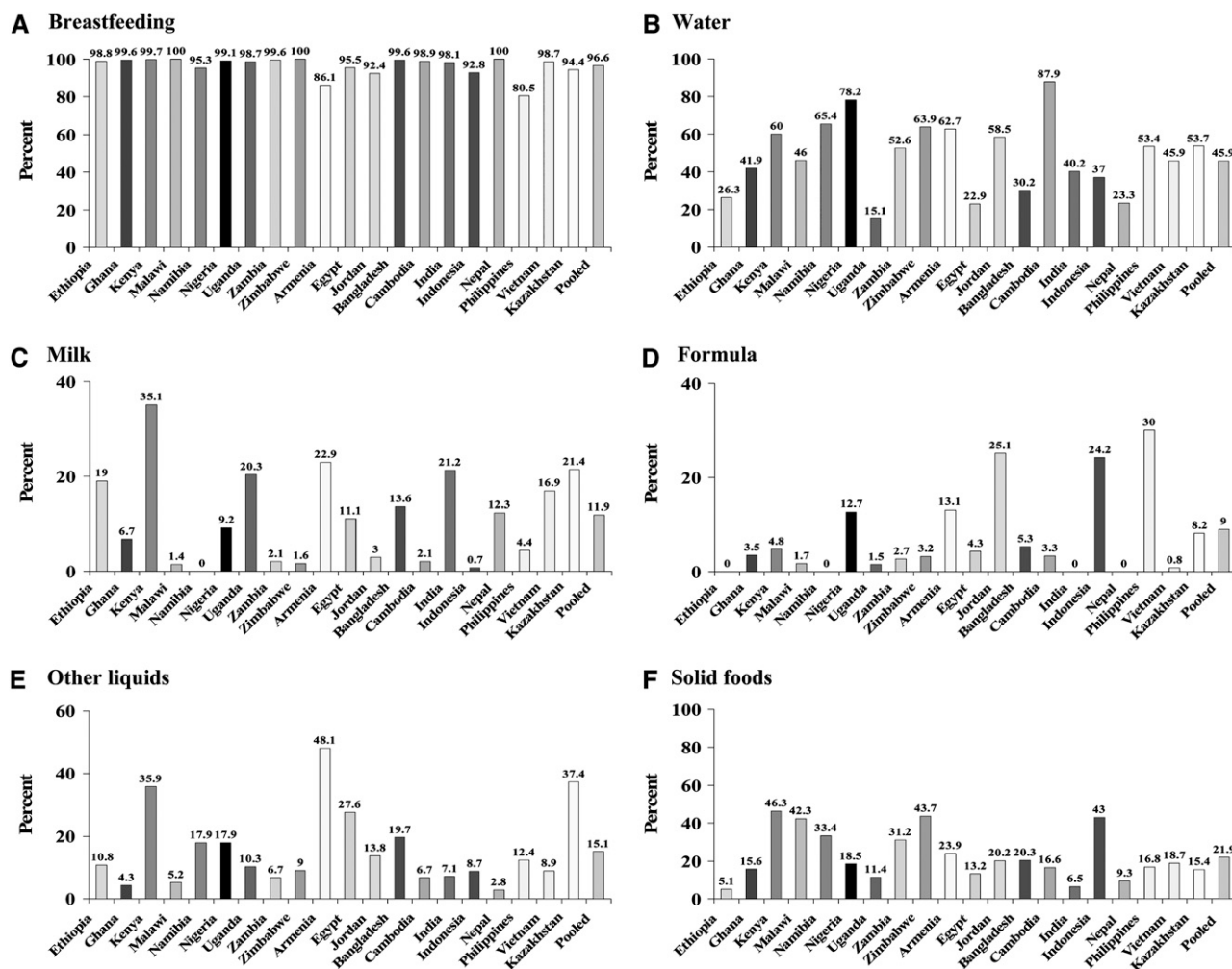


Figure 1 Percentage of mothers who stated that they currently breast-feed and separately had fed their infants 4 categories of liquid or solid food in the past 24 h by country for infants aged 0–6 mo: (A) current breast-feeding; (B) water intake; (C) other nonformula milk products; (D) infant formula; (E) other liquids; (F) solid foods.

percentages of current breast-feeding of infants aged 0–6 mo also reported the highest percentage for infants aged 6–12 mo.

The pooled percentage of mothers who reported feeding their babies water in the last 24 h was twice as high for infants at 6–12 mo as for 0- to 6-mo-old infants. This difference reflects consistently higher reported water feeding by mothers for 6- to 12-mo-old infants for all countries. Reported water feeding of the younger infants varied widely among the 20 countries, ranging from 15.1% in Uganda to 87.9% in Cambodia. Water feeding of 6- to 12-mo-old infants ranged from 65.9% in Uganda to 97.5% and 99.3% in Cambodia and Jordan, respectively.

The use of tinned, powdered, or other milks was generally low for 0- to 6-mo-old infants but was higher (as was all reported fluid feeding) in 6- to 12-mo-old infants. The variability of response was high across countries where this question was asked. In Indonesia, Malawi, and Zimbabwe, <2% of mothers responded that they used these other types of milks for infants aged 0–6 mo, whereas over one-third of mothers in Kenya reported its use for the same age group (35.1%). All countries reported a higher use of these products with 6- to 12-mo-old infants with the country-specific data showing a 1.6- to 4-fold higher use among 6- to 12-mo-olds than 0- to 6-mo infants in each country. This question was omitted in the Namibian survey instrument, so no data are available.

Sixteen of the 20 countries sampled included a specific question on commercial formula use in the last 24 h. No data are available on infant formula use for Ethiopia, Namibia, India, and Nepal. Commercial infant formula was least reported of all fluids by mothers with infants in both age categories in the 16 other countries. For the 0- to 6-mo age category, 11 of the 16 countries reported <10% formula use. The exceptions were Indonesia, Jordan, and the Philippines, in which mothers respectively reported 24.2%, 25.1%, and 30.0% use of infant formula. Mothers in these 3 countries also reported the highest use of commercial infant formula in the 6- to 12-mo age category (Jordan 28.8%, Indonesia 38.8%, and the Philippines 47.1%). Generally, reported use of formula was greater for older infants. The 2 exceptions were Zimbabwe and Armenia, where fewer mothers reported feeding their older infants formula than did mothers with infants in the younger age category (Zimbabwe: 0–6 mo, 3.2%; 6–12 mo, 2.4%; and Armenia: 0–6 mo, 13.1%; 6–12 mo, 11.6%).

As described above, use of a wide variety of other liquids was reported. A portion of these other liquids were either country- or culture-specific and had infrequently reported use. In contrast, the use of fruit juice, herbal tea, and sugar water was queried in all 20 countries. Combined, these 3 types of sweetened liquids were in widespread use. More mothers reported the use of other

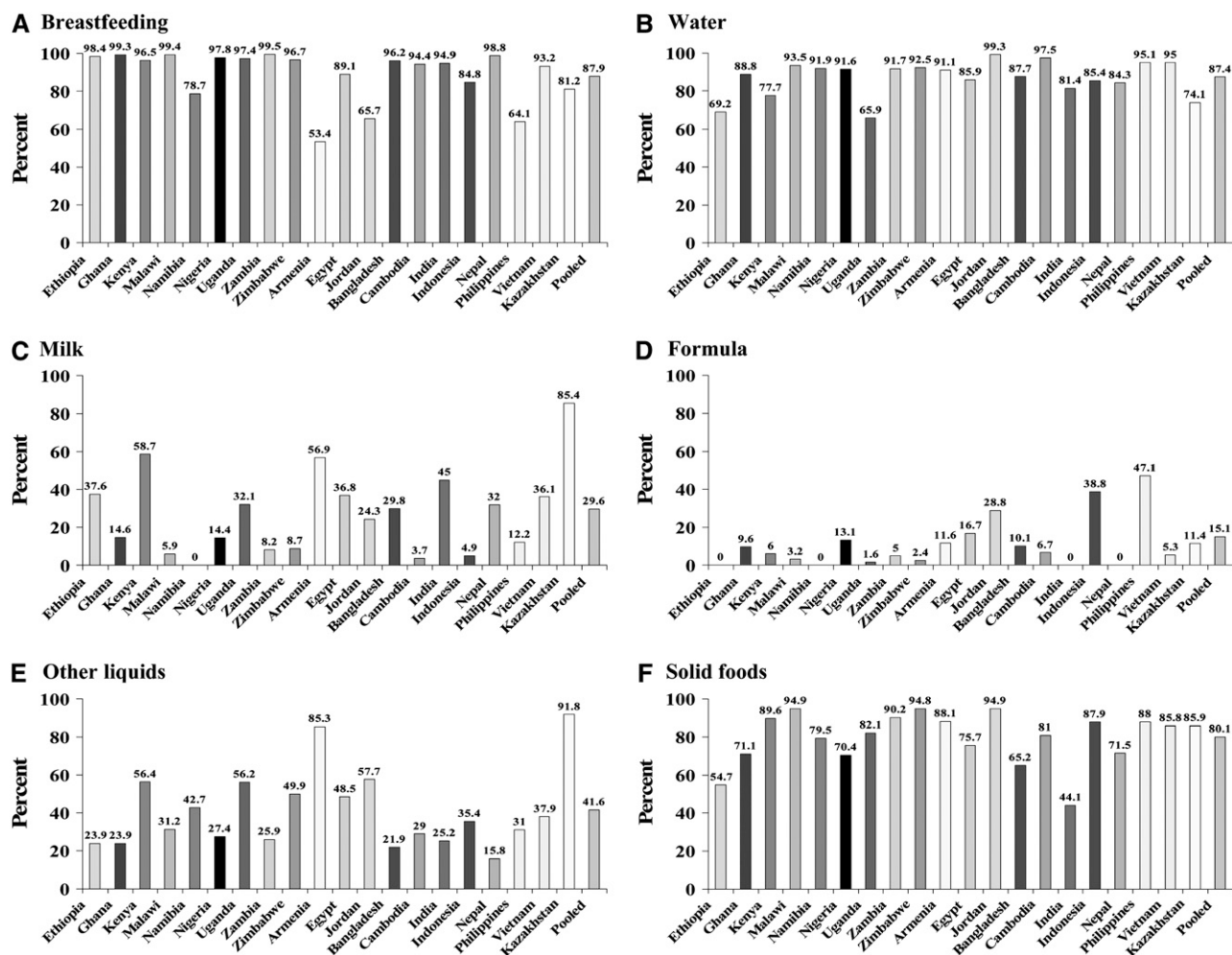


Figure 2 Percentage of mothers who stated that they currently breast-feed and separately had fed their infants 4 categories of liquid or solid food in the past 24 h by country for infants age 6–12 mo: (A) current breast-feeding; (B) water intake; (C) other nonformula milk products; (D) infant formula; (E) other liquids; (F) solid foods.

liquids than reported use of other milks or formulas. Use of other liquids ranged from 2.8% in Nepal to 48.1% in Armenia for 0- to 6-mo-old infants. Higher use was reported in all countries for the older infants, ranging from 15.8% in Nepal to 91.8% in Kazakhstan.

These pooled multicountry data indicate that over 20% of mothers are providing some type of solid food to infants aged

TABLE 1 Population-weighted averages of mothers who reported selected feeding practices during the previous 24 h for infant age groups 0–6 and 6–12 mo in 20 countries

Feeding practices	Infant age, mo	
	0–6	6–12
Percentage (weighted N)		
Current breast-feeding	96.6 (22,781)	87.9 (18,944)
Gave infant:		
Water	45.9 (10,767)	87.4 (18,663)
Tinned, powdered, or other milk	11.9 (2,769)	29.6 (6,238)
Commercial formula	9.0 (1,261)	15.1 (1,911)
Other liquids	15.1 (3,531)	41.6 (8,902)
Any solid food	21.9 (5,131)	80.1 (17,119)

0–6 mo, but the practice varies greatly among countries. Ethiopian and Indian mothers reported the lowest percentage of solid food use for their 0- to 6-mo-old infants (5.1% and 6.5%, respectively), whereas >40% of mothers in Malawi, Indonesia, Zimbabwe, and Kenya reported feeding solid foods to their 0- to 6-mo-old infants. Reported use of solid foods more than tripled to 80.1% (pooled value) for 6- to 12-mo-old infants, with reported percentages ranging from a low of 44.1% by Indian mothers to over 90% by mothers in Zambia, Zimbabwe, Malawi, and Jordan.

Discussion

For the variables presented in this article, we used weighted data and then pooled it (11) using additional weighting to account for country sample size and the number of countries who used a specific survey question. Pooling data in this fashion not only provides a much larger sample size but also creates a picture of the composite results represented by the sample countries. In this article, we present data from countries in 4 global DHS regions.

These pooled values indicate that, overall, ~97% of mothers report that they currently breast-feed their 0- to 6-mo-old infants at least some portion of the day, with 88% of mothers saying they currently breast-feed their 6- to 12-mo-old infants. These

data do not speak to the exclusivity of this practice. The DHS do not include an exclusive breast-feeding question but rather recommend using a combination of intake questions to derive an estimate of exclusive breast-feeding in a population (8). Nonetheless, the percentage of mothers of infants of both age categories who state that they are breast-feeding is encouraging in light of many recent concerns about changes in breast-feeding practices (12). In fact, the pooled mean for any-current-breast-feeding through 12 mo (nearly 88%) was substantially higher than that reported for the sample of mothers selected for study in the construction of WHO growth curves, 68% (13). This suggests that other aspects of feeding recommendations, such as the type and timing of other foods introduced, should be the focus of feeding interventions.

Pooled values indicated that mothers in these countries reported using commercial formula less than the 3 other categories of liquids to feed their babies and most often reported using plain water or other liquids with all infants 0–12 mo old. Large differences were seen among countries in mothers' use of these 4 categories of liquids. However, we are concerned by the high percentage of mothers in each country and overall who reported feeding their infants the "other liquids" category because this category consists of nearly all sweetened beverages. We expect that many of these differences will be related to other variables such as family economic status, maternal education, and household sanitation when we complete our ongoing modeling analysis (14,15).

Solid food feeding, based on pooled analysis, was higher than expected for both age categories with mothers reporting feeding solid food to 21.9% of infants 0–6 mo of age and 80.1% of infants aged 6–12 mo. These results were variable across the 20 countries. WHO recommends introduction of solids at 6 mo (16); our data show that ~20% of infants in the 20 developing countries in our sample are not compliant with this recommendation. As mentioned earlier, the DHS includes questions on specific food types that we have combined into the category of any-solid-food for the purpose of this analysis. Our future analyses will evaluate this solid food intake by country and by food type including the relation to socioeconomic variables. Based on the research reported by Dr. Krebs in this supplement, we are particularly interested in evaluating the use of meat and fish in 6- to 12-mo-old infants in this 20-country sample as well as other complementary food types and their relation to infant health (17).

To date, we have initially examined only the 24-h intake data. Ongoing analyses include data from the 7-d recalls and feeding practices for the infant's first 3 d of life. For the preliminary analysis we combined solid food into 1 category, but we will be examining specific food types and the timing of their introduction by month across the first 12 mo of life. Maternal and infant demographic and socioeconomic variables such as maternal education, maternal employment, and household variables, such as urban or rural location, water sources, the availability of refrigeration, and cooking fuels will be examined in relation to outcome variables of infant health. We are using a series of modeling techniques to elucidate the relation among these distal, intermediate, and proximate variables on infant health outcomes in these 20 countries as well as how actual practices in developing countries reflect international guidelines for infant feeding.

Question and answer session

[Q1]: The term "current breast-feeding" means any breast-feeding and does not necessarily indicate exclusive breast-feeding for 0- to 6-mo-old infants, correct? So, this extraction of

DHS data does not indicate if women are meeting guidelines for breast-feeding.

[Dr. Marriott]: Correct. The question posed by DHS is "Are you currently breast-feeding?" In the DHS there isn't an exclusive-breast-feeding question in the model questionnaire for 1999–2003. A variable can be derived from available data to provide an estimate of exclusive breast-feeding, which is something we intend to do.

[Q2]: I noticed that there are no countries included from Latin America. Is that by design or by chance?

[Dr. Marriott]: That's by design, because there were no countries in Latin America with a DHS questionnaire for the period 1999–2003 with both 24-h and 7-d intake questions that was also in English. When the country creates their country-level report and makes the data available, the questionnaire is attached. In order for us to pool the data, we needed to read each question and each variable response option and be able to verify the coding and the recoding for programming. One of our next phases is to look at the Spanish-speaking countries.

[Q3]: I come from Kenya and I've had some experience in data collection at household levels. How did you confirm that the data were obtained for/from the mothers and not for/from the primary caregiver? Given the high prevalence of HIV and AIDS in some of these developing countries, sometimes the child is in a different place than the mother.

[Dr. Marriott]: ORC Macro works very closely with the in-country survey collectors on the survey questions and methods for conducting the surveys. They also perform quality control measures before and after each survey is conducted. Survey data are not released unless they are sure they represent the design for the survey. This level of quality control is one of the reasons why these data are excellent to use, and another reason is that it allows for cross-country comparisons.

[Q4]: When you work with the data concerning diarrhea, how will you correct for instances where the family has reacted by reducing the child's food intake?

[Dr. Marriott]: There are actually DHS questions that address this specific topic, asking, "If your baby had diarrhea," and there are different time periods that this question is asked around: "Did you reduce the food intake for your baby?" "Did you provide different fluids?" And specific potential responses are elicited. So these data are available.

[Q5]: WHO recommends the method that DHS has used to derive and define "exclusive breast-feeding," in terms of excluding other liquids and foods to arrive at a derived variable. When mothers are directly asked whether they are exclusively breast-feeding, they can interpret this question differently. There has been the experience in some DHS countries where this question was asked and the findings were erroneous. One other comment, these data are cross-sectional, so there is no evidence of causality. And it's difficult to control for many aspects related to socioeconomic status that really affect both infant feeding practices and affect risk of different outcomes, be it morbidity or growth failure. I remember some early analyses of DHS by Shea or perhaps Rutstein showing that, if I remember right, breast-feeding was associated with greater outcome of diarrhea. So, this was indicative of the poorer mothers, who were breast-feeding more, under conditions in which their babies were more at risk of diarrhea. So, just a word of caution, as you look at your data, to be critical of the associations you find.

[Dr. Marriott]: Yes. I'm aware of the limitations with these approaches. With the 1999 to 2003 DHS+ dataset, ORC Macro has had trouble with their "exclusive-breast-feeding" derivation and analysis. They applied the same algorithm that they had

always used, but for some reason “exclusive breast-feeding” appeared to change dramatically in this time period. As a result, they are conducting a more in-depth analysis of the DHS+ data to see if changes in the order in which the questions were asked could have affected the results. The DHS+ 1999 to 2003 survey instruments were the first to include both 7-d and 24-h liquid and food intake questions. ORC Macro is considering whether the addition of the 7-d recall questions may have confounded their “exclusive-breast-feeding” derivation. So they have cautioned us against moving ahead with the “exclusive-breast-feeding” derivation until they reach a conclusion about their derivation process for this dataset.

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